

Calculations

$$\begin{aligned}
 \textcircled{3} \quad (1) \quad l_s &= h_s = 140 \text{ mm} \\
 \min(\theta_s) &= \arcsin\left(\frac{h_E}{l_s}\right) \\
 &= \arcsin\left(\frac{90}{140}\right) \\
 &= 40.01^\circ
 \end{aligned}$$

$$\begin{aligned}
 (2) \quad l_E &= R_o - l_s \cos(\min(\theta_s)) \\
 &= 220 - 140 \cos(40^\circ) \\
 &= 112.75 \text{ mm} \\
 \max(\theta_E) &= \arcsin\left(\frac{h_{WR}}{l_E}\right) \\
 &= \arcsin\left(\frac{65}{112.75}\right) \\
 &= 35.20^\circ
 \end{aligned}$$

$$(5) \quad R_i = l_E \cos(\max(\theta_E)) = 92.13 \text{ mm}$$

$$\begin{aligned}
 (7) \quad l_w &= 85 \text{ mm} \\
 L &\simeq 2l_w + \frac{\pi}{2}(d_2 + d_1) + \frac{(d_2 - d_1)^2}{4l_w} \\
 &\simeq 2(85) + \frac{\pi}{2}(46.57 + 15.52) + \frac{(46.57 - 15.52)^2}{4(85)} \\
 &\simeq 270.37 \text{ mm}
 \end{aligned}$$

choose $L = 268.22 \text{ mm}$

$$(8) \quad X = L - \pi \frac{(d_2 + d_1)}{2} = 268.22 - \pi \frac{(46.57 + 15.52)}{2} = 170.69 \text{ mm}$$

$$l_w = \frac{X + \sqrt{X^2 - 2(d_2 - d_1)^2}}{4} = 83.91 \text{ mm}$$

$$(10) L \simeq 2l_s + \frac{\pi}{2} (d_2 + d_1) + \frac{(d_2 - d_1)^2}{4l_s}$$

$$\simeq 2(140) + \frac{\pi}{2} \cdot 2(19.40) + 0$$

$$\simeq 340.95 \text{ mm}$$

choose $L = 335.28 \text{ mm}$

$$(11) X = L - \pi \frac{(d_2 + d_1)}{2} = 335.28 - \pi \frac{2 \cdot (19.40)}{2} = 274.33 \text{ mm}$$

$$l_s = \frac{X + \sqrt{X^2 - 2(d_2 - d_1)^2}}{4} = 137.17 \text{ mm}$$

$$(15) L \simeq 2l_E + \frac{\pi}{2} (d_2 + d_1) + \frac{(d_2 - d_1)^2}{4l_E}$$

$$\simeq 2(112.75) + \frac{\pi}{2} \cdot 2(19.4) + 0$$

$$\simeq 286.45 \text{ mm}$$

choose $L = 284.48 \text{ mm}$

$$(16) X = L - \pi \frac{(d_2 + d_1)}{2} = 284.48 - \pi \frac{2 \cdot (19.40)}{2} = 223.53 \text{ mm}$$

$$l_E = \frac{X + \sqrt{X^2 - 2(d_2 - d_1)^2}}{4} = 111.77 \text{ mm}$$

① → torque

$$\theta_{\max} = \max(\pi - (\theta_E + \theta_S)) = \max(180^\circ - (0^\circ + 40^\circ)) \\ = 140^\circ.$$

$$l_0 = \sqrt{l_S^2 + l_E^2 - 2l_S l_E \cos \theta_{\max}} \\ = \sqrt{(137.17)^2 + (111.77)^2 - 2(137.17)(111.77)^2 \cos 140^\circ} \\ = 234.09 \text{ mm} = 0.23 \text{ m}$$

$$M_0 = A(l_S + l_E)\rho \\ = (0.99 \times 10^{-3})(137.17 + 111.77)(2.7 \times 10^3) \\ = 665.42 \text{ g} = 0.67 \text{ kg}$$

$$T_S = \eta l_0^2 \left(M + \frac{M_0}{3}\right) \\ = (2\pi)(0.23)^2 \left(0.3 + \frac{0.67}{3}\right) \\ = 0.17 \text{ Nm}$$

$$\text{since } R_0 = 220 \text{ mm} \leq l_0 = 234.09 \text{ mm},$$

$$T_g = g l_0 \left(M + \frac{M_0}{2}\right) \\ = (9.81)(0.23) \left(0.3 + \frac{0.67}{2}\right) \\ = 1.43 \text{ Nm}$$

$$\rightarrow T_a > \frac{T}{i_S} = \frac{0.17 + 1.43}{3} = 0.53 \text{ Nm}$$

$$\rightarrow i_G T_M > \frac{T}{i_S} \rightarrow i_G > \frac{0.53}{0.16} = 3.3$$

→ positioning error

•> radial

$$\text{choose } T_a = 0.56 \text{ Nm}, i_G = 10$$

$$\Delta\theta \leq \frac{0.2 i_G i_S i_E}{l_S i_E \max(\sin \theta_S) + l_E i_S \max(\sin \theta_E)} \cdot \frac{180^\circ}{\pi} \\ = 1.7^\circ \times$$

choose $T_a = 0.8 \text{ Nm}$, $i_G = 18$

$$\Delta\theta \leq \frac{0.2 i_G i_S i_E}{l_S i_E \max(\sin\theta_S) + l_E i_S \max(\sin\theta_E)} \cdot \frac{180^\circ}{\pi}$$

$$= 3.1^\circ \checkmark$$

•> height

$$\Delta\theta \leq \frac{0.2 i_G i_S i_E}{l_S i_E \max(\cos\theta_S) + l_E i_S \max(\cos\theta_E)} \cdot \frac{180^\circ}{\pi}$$

$$= 2.9^\circ \checkmark$$

•> tangential

$$\Delta\theta \leq \frac{0.2 i_W i_G}{l_E \max(\cos\theta_E) + l_S \max(\cos\theta_S)} \cdot \frac{180^\circ}{\pi}$$

$$= 2.9^\circ \checkmark$$

Summary

$$R_o \approx 220 \text{ mm}, h_e \approx 90 \text{ mm}, h_{WR} = 65 \text{ mm}, h_s = 140 \text{ mm}$$

① Stepper motor ← from appendix 17

$$(1) \Delta\theta = 1.8^\circ$$

$$(2) T_a = 0.56 \text{ Nm}$$

$$(3) f_o = i_w i_g \omega / \Delta\theta$$

$$= (3)(10) \left(\frac{\pi}{12} \cdot \frac{180^\circ}{\pi} \right) \cdot \left(\frac{1}{1.8^\circ} \right)$$

$$= 250 \text{ Hz}$$

$$(4) t_o = 1000 \Delta\theta / (i_w i_g \eta) = 0.093 \text{ s}$$

$$(5) \omega_{WR} = (f_o \Delta\theta / i_{WR} i_g)$$

$$= (250) \left(1.8 \cdot \frac{\pi}{180} \right) \cdot \left(\frac{1}{2} \right) \cdot \left(\frac{1}{18} \right)$$

$$= \frac{5\pi}{72} \frac{\text{rad}}{\text{s}}$$

$$(6) \omega_H = (25/26) \omega_{WR} = \frac{5\pi}{36} \frac{\text{rad}}{\text{s}}$$

Model number of selected stepper motor (reduction ratio i_g):

CSK 243AP-SG18 ($i_g = 18$) ← from appendix 17

② End effector ← from appendix 19

(7) Air gripper: Operation range 24.9 ~ 30.9 mm

Air pressure 0.5 MPa

Maximum grasping force 27 N

Weight 0.115 kg

Model number of selected end effector:

MHZ2-16S ← from appendix 19

③ Design Parameters

a. Operation range of each joint

- (1) $40^\circ \leq \theta_s \leq 90^\circ$ ←
 - (2) $0^\circ \leq \theta_E \leq 35^\circ$ ←
 - (3) $-90^\circ \leq \theta_W \leq 90^\circ$ ←
 - (4) $-30^\circ \leq \theta_{WR} \leq 30^\circ$ ←
 - (5) $R_i = 92.13 \text{ mm}$ ←
- from requirement

b. Timing pulley and timing belt for waist joint ($i_w = 3$)

(6) $z_1 = 24, d_1 = 15.52 \text{ mm}, z_2 = 72, d_2 = 46.57 \text{ mm}$

Model number: $\begin{cases} z_1: \text{ATP24MXL025} \leftarrow \\ z_2: \text{ATP72MXL025} \leftarrow \end{cases}$

(7) $L = 268.22 \text{ mm}, b = 6.4 \text{ mm}$ (# of tooth: 132) ← from appendix 15

Model number: TBN132MXL025

(8) $l_w = 83.91 \text{ mm}$ ←

c. Timing pulley, timing belt and gear for shoulder joint and elbow joint. ($i_s = i_E = 3$)

(9) $m = 0.8 \text{ mm}, \alpha = 20^\circ, z_1 = 30, z_2 = 90, b = 5 \text{ mm}$

Model number: $\begin{cases} z_1: \text{S80SU30B+0505} \leftarrow \\ z_2: \text{S80SU90B+0508} \leftarrow \end{cases}$ from appendix 15 (spur)

(10) $L = 335.28 \text{ mm}$ (# of tooth: 165) ← from appendix 15

Model number: TBN165MXL025

(11) $l_s = 137.17 \text{ mm}$ ←

d. Timing pulley, timing belt and gear for hand and wrist joint ($i_{WR} = 2$)

(12) $m = 0.5 \text{ mm}$, $\alpha = 20^\circ$, $z_1 = 40$, $z_2 = 80$,

$b_1 = 5 \text{ mm}$, $b_2 = 2 \text{ mm}$

Model number: $\begin{cases} z_1: \text{SS0540B+0505} \leftarrow \\ z_2: \text{SS0B80A-0208} \leftarrow \end{cases}$ from appendix 13 (spur)

(13) $m = 0.8 \text{ mm}$, $\alpha = 20^\circ$, $z_1 = 20$, $z_2 = 40$

Model number: $\begin{cases} z_1: \text{B80B20} \leftarrow \\ z_2: \text{B80B40} \leftarrow \end{cases}$ from appendix 12 (bevel)

(14) $z_1 = z_2 = 30$, $d_1 = d_2 = 19.4 \text{ mm}$

Model number: $\text{ATP30MXL025} \leftarrow$

(15) $L = 284.48 \text{ mm}$ (# of tooth: 140) $\leftarrow$ from appendix 15

Model number: $\text{TBN140MXL025} \leftarrow$

(16) $\lambda_E = 111.77 \text{ mm} \leftarrow$